AutoEncoder-Compressed Parallel Split Learning for Pre-trained Model Fine-Tuning

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01 Context & approach

The challenge

Parallel split learning offloads model computation but exchanges activations and gradients at every step. Compressing pretrained features with an unaligned autoencoder can destroy task accuracy. *AE-PSL* aligns the compressor before fine-tuning to reduce communication while preserving useful representations.

Our contribution

Compress activations and gradients through a lightweight split-layer autoencoder.
Align to public pretrained features, then adapt to each client's distribution.
Evaluate global and local inference across four vision benchmarks.

1 Align to pretrained features

Train the autoencoder on public ImageNet100 features with the pretrained model frozen.

2 Adapt to each client

Warm up locally, retain client-specific encoders, and average decoders into one shared server decoder.

3 Fine-tune with compression

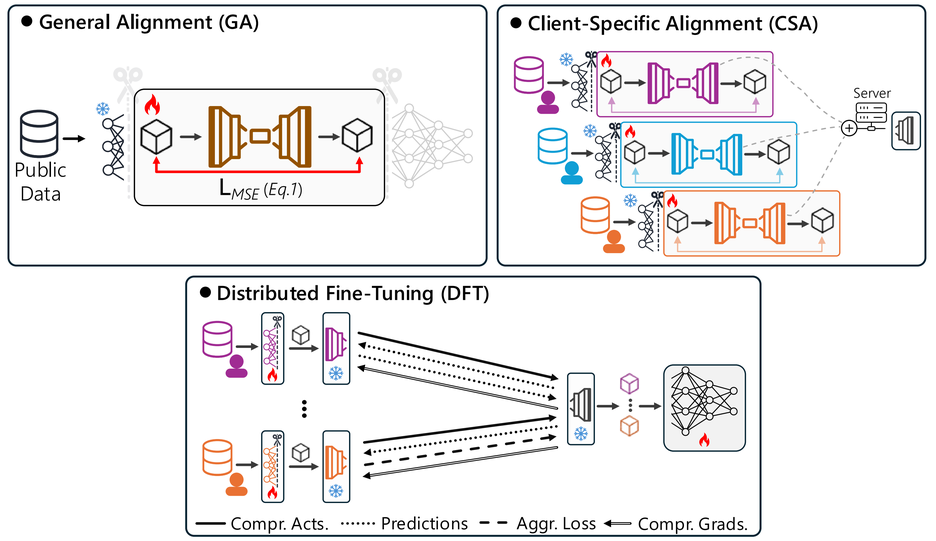
Freeze the autoencoder; exchange compressed features and gradients. Classification tokens bypass compression; labels remain on clients.

Scope

Validated for supervised ViT-B/32 vision fine-tuning; multimodal and language adaptation remain untested. Requires public-data alignment and local warm-up; global and local inference use different compression settings.

02 Method

Align learned compression with pretrained features before distributed fine-tuning.



Core formulation

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} \|H(x) - D_\psi(E_\phi(H(x)))\|_2^2$$

H: frozen pretrained features; E/D: encoder/decoder. Minimize reconstruction error on public data, then each client's data.

Paper

Code

03 Results

≈10×
communication reduction
Relative to uncompressed split fine-tuning.

82.7 ± 0.2%
CIFAR-100 accuracy
5 clients; matches the uncompressed reference.

Accuracy at approximately 10× communication reduction

Method	CIFAR-100	Food101	SUN397	FEMNIST
No compression (R=1)	82.7 ± 0.2	69.1 ± 0.9	67.0 ± 0.6	87.9 ± 0.3
C3-SL	50.9 ± 1.2	40.3 ± 1.2	41.3 ± 2.1	83.5 ± 0.1
Rand-Top-K	76.2 ± 0.4	60.2 ± 0.2	62.9 ± 1.0	87.2 ± 0.4
ADC	68.4 ± 0.4	54.6 ± 1.6	52.9 ± 1.1	81.1 ± 0.9
AE-PSL (Ours)	**82.7 ± 0.2**	**69.6 ± 1.2**	**66.7 ± 0.3**	**88.0 ± 0.1**

Top-1 accuracy (%), mean ± SD, 3 seeds. ViT-B/32; 5 clients; compressed methods R≈10. First three datasets: global inference, AE bypassed. FEMNIST: local inference, AE active.

Alignment prevents feature mismatch

Ablation	CIFAR-100	FEMNIST
Unaligned AE	1.4 ± 0.3	86.9 ± 2.0
+ General alignment	75.2 ± 0.4	87.1 ± 1.1
+ Freeze AE	79.6 ± 0.9	88.0 ± 0.6
+ Client alignment (Ours)	**82.2 ± 0.7**	**88.0 ± 0.7**

Accuracy (%), reported ± values, 3 seeds; ViT-B/32, 5 clients, R=15. CIFAR-100: global; FEMNIST: local inference. Cumulative ablations; final row is *AE-PSL*.

Takeaway

AE-PSL preserves uncompressed CIFAR-100 accuracy at **≈10× lower communication**, while outperforming the tested compression baselines. Alignment and freezing are crucial; global inference bypasses the autoencoder, whereas local FEMNIST inference retains it.